Soft Shells: Carbon-Absorbing Concrete Structures through Fabric Formwork

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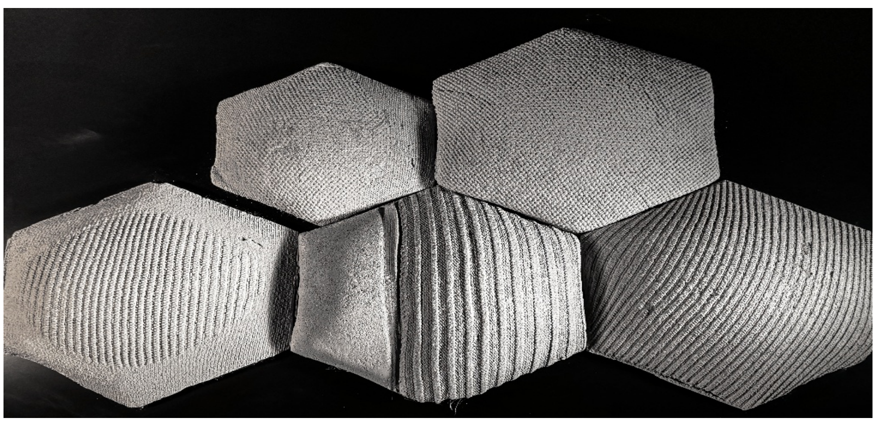
ABSTRACT: Concrete remains one of the most environmentally impactful materials in global construction, however, emerging strategies can mitigate its footprint and enable carbon-sequestering elements. *Soft Shells* explores the integration of controlled knitted fabric formwork with CO_2-sequestering concrete mixes to develop low-carbon, high-performance architectural shells. Stretchable weft-knit textiles are designed to deform predictably under the hydrostatic pressure of wet concrete. By adjusting local material properties through CNC knitting and localized reinforcement, the fabric guides curvature formation, generating curved geometries that are structurally efficient and formally expressive. These geometries increase exposed surface area, enhancing CO_2 absorption during and after curing. Fabric formwork further improves sustainability through reduced material use, lightweight construction, and reusability. To evaluate environmental performance, three additive strategies were tested: diatomaceous earth, titanium dioxide (TiO_2), and fly ash. Concrete shell specimens incorporating each additive were compared with control mixes under controlled CO_2 monitoring. Results confirm that surface-area amplification through fabric-formed geometry, combined with targeted additive incorporation, measurably improves post-cure carbon sequestration. This research proposes a locally implementable and globally relevant framework for reducing embodied carbon while expanding the formal and environmental potential of concrete architecture.

KEYWORDS: carbon-absorbing concrete, fabric formwork, sustainable construction, material efficiency
PAPER SESSION TRACK: Technologies of Place - Localized Innovation for Global Impact



INTRODUCTION

The search for sustainable construction methods has driven a paradigm shift in how materials are conceived, fabricated, and deployed. Textile-based formwork offers an adaptable and low-impact method for shaping concrete, offering a new formal grammar and vocabulary for architectural shells. Knitted textiles possess intrinsic mechanical versatility due to their looped architecture, which enables controlled deformation and anisotropic elasticity. These characteristics have long been exploited in technical and biomedical applications, yet their potential within architectural contexts remains underexplored. When integrated as flexible formwork, knitted fabrics can translate soft, adaptive behavior into rigid structural outcomes, allowing geometry to emerge naturally under gravity and pressure rather than through rigid molds or extensive formwork systems. The Soft Shells project extends this textile logic to the realm of sustainable concrete construction. The research investigates how digitally designed knitted formwork can be paired with carbon-absorbing concrete composites to create low-carbon, high-performance shell structures. The research combines three sequestration strategies: diatomaceous earth, titanium dioxide (TiO_2), and fly ash, to enhance CO_2 uptake during long-term exposure. By coupling material science with computational fabrication, this study demonstrates how surface-active geometries and additive-rich mixes can transform concrete into an active participant in carbon capture. The broader goal is to propose a new design paradigm for concrete shells that minimizes environmental impact.

1.0 BACKGROUND

Cement is the most widely used construction material, but it also represents one of the primary sources of anthropogenic carbon emissions, contributing approximately 7% of total global CO_2 emissions (Anderson and Moncaster 2020). The construction industry consumes nearly two-fifths of all raw materials extracted from nature, including sand, gravel, limestone, metals, and water (Valentini 2023), much of which is used for the production of concrete. The carbon footprint of the construction industry is expected to double by 2050 (Li et al. 2025). There is a need to find alternative design and construction methods that reduce the climate impact compared to traditional methods, helping to address global emission problems. Recent advances in concrete casting have introduced textile-based formwork systems as an alternative to conventional rigid formwork (Veenendaal and Block 2014). The use of flexible fabrics is notable for its ability to generate light, reusable, and resource-conscious molds capable of producing intricate architectural forms (Orr 2019). When filled with liquid concrete, the textile surface conforms under gravitational and hydrostatic forces, shaping itself into naturally curved catenary geometries that are both aesthetically refined and structurally efficient, while significantly lowering material demand and fabrication expenses compared to plywood or steel counterparts (West 2017). Membrane deformation for shell optimization has been studied for woven or composite formwork with minimal stretch (Popescu et al. 2018; Fang 2009). However, knitted fabrics present a more challenging design situation due to their low stiffness and high deformability. Knit structures, particularly weft knits, can accommodate the stretch required for high-strain applications. Araujo (2011) describes the ability of weft knit fabrics to perform under load conditions due to their orthotropic mechanical properties. Compared to woven fabrics, knits offer a significantly higher stretch capability, reaching up to 100% compared to 10% for woven fabrics (Cooke 2011). This is due to the yarn geometry of the knit fabric, which includes loops that can change shape before significant yarn loading is realized (Vassiliadis et al. 2007). Gil Pérez et al. (2016) developed nonlinear analytical methodologies for membrane and tensile fabric systems, exploring how deformation, stiffness, and load-bearing capacity of knitted fabrics are governed by stitch topology, inter-loop kinematics, and yarn properties. In addition to advancements in textile formwork, significant research has been directed toward carbon-absorbing or photocatalytic concrete composites. Additives such as diatomaceous earth, fly ash, and titanium dioxide (TiO_2) have been shown to enhance CO_2 capture, improve surface reactivity, and reduce clinker content in cementitious systems (Kipsanai et al. 2022).

1.1 Scope of Research

While these strategies address chemical sequestration, few studies have investigated their interaction with surface morphology and geometry, factors that strongly influence the kinetics of CO_2 diffusion. Considering these developments, the Soft Shells research project advances a synthesis of textile engineering, material science, and sustainable architecture. By employing digitally controlled knitted formwork in conjunction with carbon-sequestering concrete mixtures, the study aims to demonstrate how form-active geometry and materials science can work synergistically to produce low-carbon, high-performance concrete shells. The scope is limited to laboratory-scale investigations that examine how stitch geometry and boundary tension influence form under the hydrostatic pressure of fresh concrete. Four distinct knit structures (stockinette, garter, rib, and seed) were produced and evaluated experimentally and through nonlinear finite element simulations to model the cured shell geometry. Parallel to the formwork analysis, the research examines the role of three additives for carbon sequestration: diatomaceous earth, titanium dioxide (TiO_2), and fly ash. These additives were incorporated into controlled cementitious mixes to assess post-cure CO_2 absorption under standardized temperature and humidity conditions. The study focuses on surface carbonation and form-induced geometric effects, rather than bulk mineral carbonation or long-term structural degradation. The work is confined to small-scale prototypes and simulated models that replicate the behavior of flexible formwork systems at the architectural component scale. Field-scale implementation, long-term environmental exposure, and full structural load testing lie outside the present scope but are identified as future research directions. The overall intent is to establish a computational and experimental framework for integrating mechanically active textiles with sustainable concrete technologies, providing foundational data for future development of low-carbon, textile-informed architectural structures.

1.2 Objective

The primary objective of this research is to develop and evaluate a textile-integrated framework for producing low-carbon concrete shells using knitted formwork and carbon-absorbing cementitious mixes. Specifically, the research seeks to:

i. Characterize the mechanical response of knitted textiles, including stockinette, garter, rib, and seed structures, under tensile loading in course and wale directions to determine their suitability as flexible formwork materials.
ii. Simulate fabric deformation using nonlinear finite-element analysis to predict displacement, stress distribution, and curvature formation during concrete casting. Models are validated with physical prototypes.
iii. Assess the impact of additive composition, including diatomaceous earth, titanium dioxide (TiO_2), and fly ash, on CO_2 absorption capacity and post-cure surface reactivity of concrete shells formed with textile molds.
iv. Propose a scalable methodology that integrates mechanical testing, digital simulation, and mix design as a basis for future development of textile-based, carbon-responsive architectural systems.
v. Propose a modeling method that enables the designer to choose appropriate knitted structures for achieving desired shell geometry.

2.0 METHODOLOGY

This research employed a hybrid experimental and computational methodology, combining textile testing, fabric-formed concrete casting, and finite-element simulation, to evaluate the mechanical and environmental performance of knitted formwork systems. The experimental component focused on quantifying the anisotropic elastic behavior of knitted fabrics through tensile testing and controlled casting using formwork. The mechanical data obtained from these experiments served as input parameters for a nonlinear finite-element analysis (FEA) in SolidWorks Simulation, enabling the prediction of stress distribution, displacement, and curvature development in the fabric under realistic loading conditions. Parallel to the mechanical evaluation, a series of CO_2-absorption experiments was performed on concrete shells prepared with different additive formulations, including fly ash, diatomaceous earth (DE), and titanium dioxide (TiO_2), to measure the influence of material composition and surface geometry on post-cure carbon uptake. This dual approach provided a feedback loop between physical testing and digital modelling, validating the experimental deformation data and confirming the simulation results, while computational predictions informed adjustments in the casting and fabric-tensioning procedures. The methodology thus bridges material science, textile engineering, and architectural design, establishing a reproducible framework for developing low-carbon concrete systems guided by textile behaviors and environmental performance.

2.1 Materials

Four knitted fabric structures - stockinette, garter, rib, and seed - were fabricated using acrylic yarns with uniform gauge and thickness. Each fabric was characterized by its anisotropic mechanical properties along the course (x-direction) and wale (y-direction). Concrete mixtures were prepared using Ordinary Portland Cement (OPC) and Polymer-Modified Cement (PMC) with a constant water-to-cement ratio (W:C:S). Three additive types - diatomaceous earth, fly ash, and titanium dioxide (TiO_2) - were introduced separately to investigate their effects on CO_2 absorption and surface reactivity. The fabrics were produced using acrylic yarns on a Shima Seiki SES-SWG, a flatbed computerized weft knitting machine. These four different fabrics were used as formwork for concrete shells, although numerous additional knit types and patterns could be produced. The types of stitches used are explained below and shown in **Figure 1**. Links-links is the flat-bed knitting industry term for seed and garter structure. (Garter and Seed Stitch are hand-knitting terms). Knitting is on a flat-bed electronic knitting machine (as opposed to a circular or warp knitting machine). Links-links indicate the usage of two beds producing front-facing and back-facing stitches in a defined pattern. Single jersey or Stockinette is created on one bed of a flat-bed machine and consists of only front-facing knit stitches. Rib, seed, and garter stitches are variations of Links-links and use two beds of a flat-bed knitting machine. Garter consists of one course of front stitches, followed by one course of back stitches and repeated throughout. Rib consists of vertical wales of front or back stitches. Seed stitch consists of a checkerboard (for foursquare) formation of front and back stitches in repeat.

i) Stockinette: This structure consists entirely of knit stitches arranged in a uniform sequence. Due to its loop orientation, the fabric exhibits moderate elasticity, with greater stretch along the width (course direction) than along the length (wale direction).
ii) Garter Knit: Created by alternating rows of knit and purl, this pattern has alternating loop orientation which gives the fabric a balanced yet flexible structure, with higher stretch along the wale direction compared to the course.
iii) Rib Knit: Formed by alternating vertical columns of knit and purl, the rib structure generates an interlocked loop structure between adjacent columns enable the fabric high elasticity across its course direction.
iv) Seed Knit (Moss Stitch): This pattern alternates knit and purl stitches both across rows and columns, producing a texture resembling a checkerboard. The seed knit showed the greatest overall flexibility and balanced stretch.

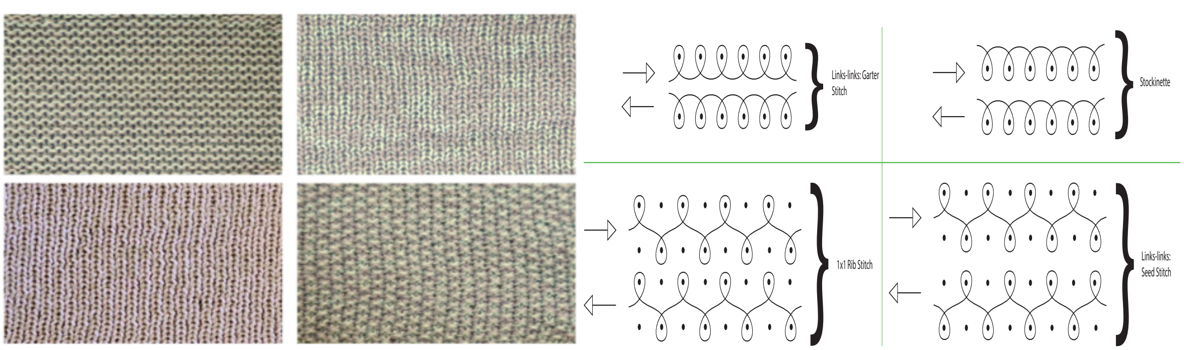


Figure 1 : Photograph of the four knit samples (left) and stitch diagram (right). In order- Top left - Garter Stitch, Top Right - Stockinette Stitch, Bottom Left - Rib Knit, Bottom Right - Seed Stitch

2.2 Simulation And Modeling Procedure

Finite-element simulations were performed in SolidWorks using a nonlinear static analysis. The fabrics were modeled as 2 mm-thick orthotropic shell bodies. Material constants, including elastic modulus in course (x-direction), in wale (y-direction) E_x and E_y respectively, were derived from tensile tests on fabric specimens. Uniform pressure loads corresponding to wet-concrete weight were applied. Boundary conditions replicated frame restraints from the experimental setup. Mesh refinement studies were conducted to optimize accuracy versus computation time. The predicted displacements and stress distributions were compared to experimental shell geometries to validate the modeling approach. The applied pressure applied was based on the average weight of concrete before and after drying to get accurate model. Table 1 depicts the solver parameters used in the simulation model. Pressure applied is as per the average weight of wet and dry concrete to have optimal criteria for long term impact of concrete on fabric deformation. To evaluate the directional stiffness of different knitted structures for use as fabric formwork, tensile test, also known as grab test (D13 Committee, n.d. 2025), was performed on each fabric type in both the course (x) and wale (y) directions. The stress-strain data obtained from the tests were used to calculate the elastic moduli (Ex and Ey)

corresponding to low strain regions (within the linear range). The results are summarized in Table 2, which compares the elastic moduli of three composite knit types, Rib-Stockinette, Seed-Stockinette, and Garter-Stockinette, each consisting of a patterned knit (rib, seed, or garter) integrated with a plain stockinette layer. The shear modulus is estimated using the formula $G_{xy} = 0.3 * \sqrt{Ex * Ey}$ for the calculated elastic modulus in the inputs for simulation (The Institute of Paper Chemistry et al. 1981) . A coefficient of 0.3 was adopted rather than the standard 0.387, reflecting the higher deformability and stretch characteristics of knitted fabrics compared to the stiffer paper samples used in the original empirical correlation.

Table 1 : Solver Parameters used for different types of knits

Parameter	Rib-Stockinette	Seed-Stockinette	Garter-Stockinette
Study Name	Nonlinear 1	Nonlinear 1	Nonlinear 1
Analysis Type	Nonlinear-Static	Nonlinear-Static	Nonlinear-Static
Mesh Type	Shell (Surface)	Shell (Surface)	Shell (Surface)
Shell thickness (mm)	2	2	2
Nodes	25,973	29,293	9,791
Elements	12,670	14,292	4,728
Pressure (Pa)	204.2	138.5	160.5

Table 2 : Material Properties of the knit fabric used in the Finite Element Analysis

Property	Rib-Stockinette		Seed-Stockinette		Garter - Stockinette	
	Rib	Stockinette	Seed	Stockinette	Garter	Stockinette
Type	Orthotropic	Orthotropic	Orthotropic	Orthotropic	Orthotropic	Orthotropic
Elastic Modulus (Ex) (MPa)	0.03	0.0491	0.0764	0.0491	0.148	0.0511
Elastic Modulus (Ey) (MPa)	0.153	0.1729	0.144	0.1729	0.0664	0.1441
Poisson's Ratio	0.4	0.4	0.4	0.4	0.4	0.4
Shear Modulus (Gxy) (MPa)	0.019	0.028	0.031	0.028	0.03	0.026

2.3 Fabric Formwork Preparation

Each knit fabric was stretched between hexagonal wooden frames to create a uniform tension field and consistent boundary conditions. Clamps were used to secure the frames, forming a stable platform for concrete casting. The fabric's elasticity and loop geometry allowed controlled deformation under hydrostatic pressure, creating unique shell geometries when filled with fresh concrete. Surface deformations and curvatures were visually recorded and later compared with simulated geometries to assess model accuracy. Figure 2 below shows the apparatus for the experimental verification of the simulated concrete mixture shells. Concrete was applied to the pre-stretched fabric carefully to not put any manual pressure due to human application. The aim is to allow concrete to flow solely under gravity. Figure 3 shows the system after the concrete has been applied to the fabric.

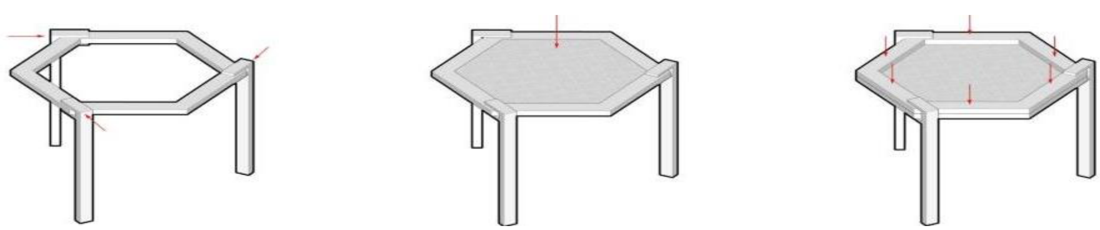


Figure 2 : In order from left to right (Drawing for wooden hexagonal frame which is constructed using lap joints and are held by F-clamps, The fabric is stretched and held between the upper and the lower frames.

Figure 3 : Fabric placed between the hexagonal frames (left), frames connected with the clamps (middle), concrete applied on the fabric (right).

2.4 Concrete Mix Design and Additives

Additive design shown in Table 3 represents the twelve concrete mixture recipes developed for casting fabric-formed shells. Two main matrix types were employed: Standard Cement (SC) and Polymer-Modified Cement (PMC). Each category incorporated four additives, Fly ash (FA), Diatomaceous Earth (DE), Titanium Dioxide (TiO_2), and Sand (SA), in various combinations to evaluate their influence on CO_2 absorption and surface formation. The SC series (Mix 1–2) included duplicate batches for each additive to ensure repeatability and consistency, while the PMC series (Mix 3) used the same additive set but with polymer modification to enhance flexibility and carbonation reactivity. This systematic design enabled direct comparison between conventional and polymer-modified systems under identical curing and testing conditions.

Table 3 : Twelve concrete mixture recipes used for shell manufacturing with additives Fly ash (FA), Diatomaceous Earth (DE), Titanium Dioxide (TiO2) , Sand (SA), Cement (CEM) and Polymer Modified Cement (PMC).

Mix Category	Additive Combinations
Standard Cement (SC)	Fly ash (Mix 1–2); Diatomaceous Earth (Mix 1–2); Titanium Dioxide (Mix 1–2); Sand (Mix 1–2)
Polymer-Modified Cement (PMC)	Fly ash (Mix 3); Diatomaceous Earth (Mix 3); Titanium Dioxide (Mix 3); Sand (Mix 3)

2.5 CO_2 Measurement Setup

Figure 4 shows the experimental setup used for monitoring CO_2 absorption in concrete shells. Each specimen was sealed inside an airtight plastic chamber along with a calibrated Aranet4 CO_2 sensor, which continuously measured the change in CO_2 concentration over time. The Aranet sensor recorded real-time CO_2 levels (ppm), temperature (°C), and relative humidity (%) at five-minute intervals. By tracking the reduction in CO_2 concentration from an initial value of around 600 ppm, the absorption rate and total cumulative uptake were calculated for each mix. This non-destructive setup ensured consistent environmental conditions, allowing accurate comparison of reaction speed and long-term CO_2 sequestration capacity among the Standard Cement (SC) and Polymer-Modified Cement (PMC) shells.

Figure 4 : The experimental setup used for monitoring CO_2 absorption in concrete shells.

3.0 RESULTS AND DISCUSSION

Figure 5 shows the concrete shells cast using different knitted fabric formworks. Each fabric pattern produced a distinct surface geometry and curvature response under the hydrostatic pressure of wet concrete. The Rib-Stockinette shell exhibits deeper transverse ridges and higher vertical stretches, reflecting the greater elasticity of the rib structure. The Seed-Stockinette shell displays a relatively uniform, balanced texture due to its alternating knit–purl configuration, providing moderate deformation in both directions. The Garter-Stockinette shell shows denser horizontal corrugation, indicating higher compliance along the course direction and enhanced stiffness along the wale direction. These variations confirm that the knit topology directly influences surface texture and form development, allowing control over curvature and stiffness in fabric-formed shells.

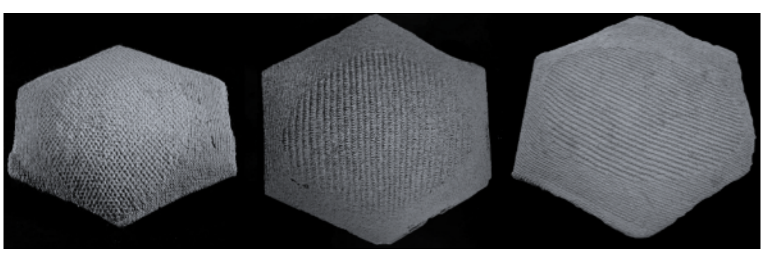


Figure 5 : Seed-Stockinette formed shell (Left), Rib-Stockinette Shell (In-the-middle) and Garter-Stockinette Shell (Right)

3.1 Comparison Between Simulated and Experimental Results

To illustrate the parity between the simulated and the experimental results, concrete shells were scanned using Creaform-Ametek Ultra Precision 3D Scanner. To compare the curvature of the shell, the simulated surface and the scanned shell surface were superimposed to match the respective surfaces, as shown in Figure 6. The legend illustrates the results of the difference in top curvature between the course and wale directions. The blue color shows that the simulated surface is above the scanned surface by 4.42mm and the orange color depicts that the simulated surface is below the scanned surface by 4.42mm. The green part validates that the simulated and scanned surfaces match over the curvature of the shell. The top and side views validate that the simulated models match the curvature of the shell within 5mm. Figure 6 illustrates that the surface of simulated shell is compared to the scanned surface at the bottom of the ridges which are due to the fabric structure for accurate comparison. Transparent shell is the scanned one for better visualization of the curvature comparison. Similarly, the process is applied to Seed-Stockinette shell as illustrated in Figure 7. The formed shell for Garter-Stockinette shell is compared with the simulated model in (Figure 8).

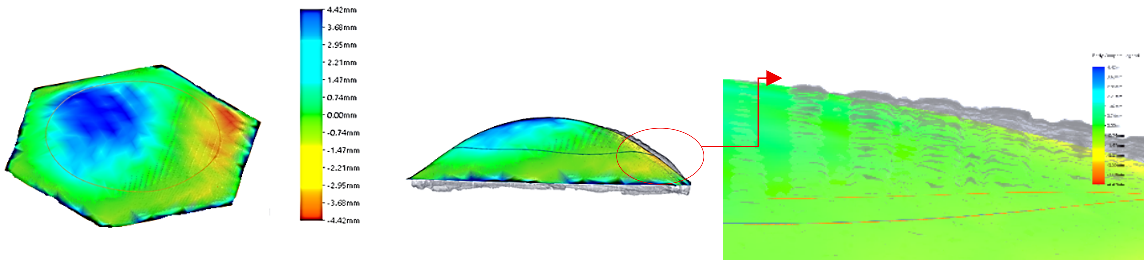


Figure 6 : Rib-Stockinette concrete shell comparison between the Simulated and Scanned models Top view (to the left) and Side view (middle) and closeup of the comparison for simulated and scanned shells (to the right)

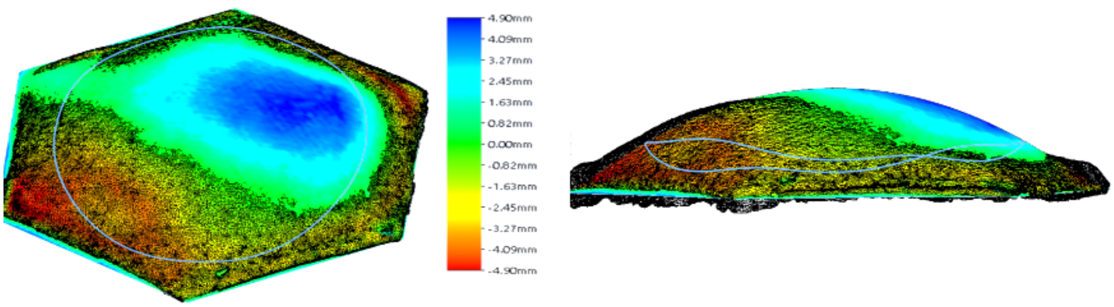


Figure 7 : Seed-Stockinette concrete shell comparison between the Simulated and Scanned models Top view (to the left) and Side view (to the right). The highlighted blue boundary is the fabric deformation in both views occurring due to the concrete load.

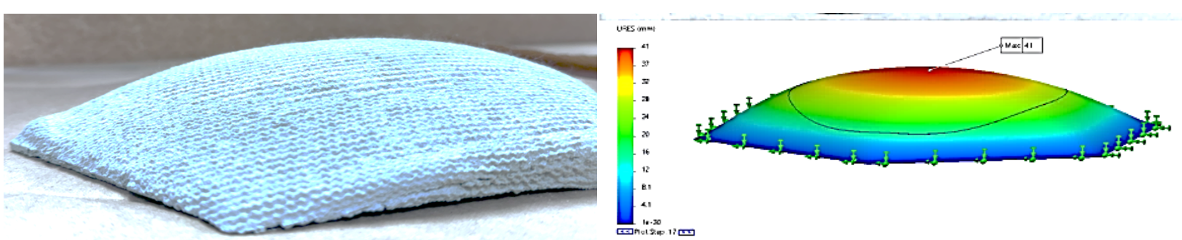


Figure 8 : Simulated model compared to actual fabricated shell for Garter-Stockinette fabric. The predicted curvature (on right) as well as the maximum deformation is within 5mm of actual shell (on left).

3.2 CO_2 Absorption and Environmental Performance

Figure 9 compares the cumulative CO_2 absorption capacity and the average time required for concentration reduction across all concrete additive mixtures. SC Fly ash exhibited the highest overall absorption (301.2 mg/g), followed closely by SC TiO_2 (284.9 mg/g) and PMC Normal (273.7 mg/g), indicating strong long-term sequestration potential. In contrast, diatomaceous mixes showed the lowest absorption due to limited reactivity. The reaction time graph shown in Figure 10 shows that PMC Fly ash and PMC TiO_2 achieved the fastest CO_2 reduction (≤2 h), demonstrating high short-term reactivity, while SC TiO_2 required ≈20 h, reflecting slower but sustained carbonation and photocatalytic absorption. Overall, PMC mixes favor rapid initial CO_2 uptake, whereas SC TiO_2 and SC Fly ash provide higher long-term absorption capacity. In Figure 11, the vertical axis represents the cumulative average CO_2 absorbed over a fixed duration of 3000 minutes, while the horizontal axis indicates the time required to achieve a specific reduction in ambient CO_2 concentration, thereby capturing the kinetic rate of absorption. Samples located toward the left side of the graph (low reaction time) indicate faster CO_2 absorption kinetics. PMC TiO_2 and PMC Normal exhibit the fastest CO_2 reduction rates, reaching the target concentration within approximately 3–5 hours while maintaining relatively high cumulative absorption values (~260–280 mg/g). PMC Fly ash also shows quick response and moderate absorption, suggesting its suitability where immediate sequestration is desired (e.g., freshly cast or thin-shell structures exposed to air). These materials exhibit enhanced short-term reactivity, likely due to their higher surface area and increased chemical reactivity at the polymer-modified interface, which promotes faster CO_2 binding. In contrast, samples positioned toward the right side (longer reaction times) but with high cumulative values indicate slower but more sustained absorption over time. SC TiO_2 takes the longest (~20 hours) to reach 300 ppm but achieves one of the highest total CO_2 absorption capacities (~300 mg/g). This behavior reflects slow, continuous carbonation and photocatalytic reaction activity, typical of TiO_2-enhanced matrices where diffusion-controlled processes dominate over immediate surface reactions. SC Fly ash exhibits both high capacity and moderate reaction speed (~6 hours), indicating a balance between rapid surface reactions due to pozzolanic activity and longer-term microstructural carbonation. SC Diatomaceous and PMC

Diatomaceous perform less effectively overall, possibly due to reduced available $Ca(OH)_2$ content or limited surface reactivity of the diatomaceous additive.

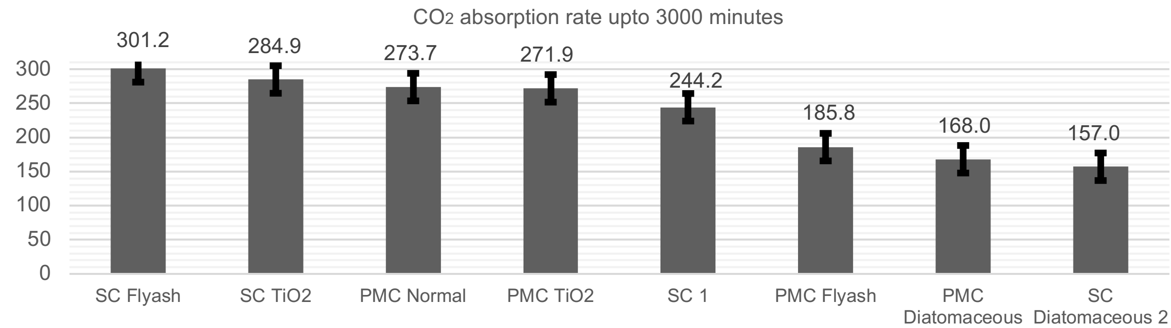


Figure 9 : Bar graph in descending order of absorption rate up to 3000 minutes for the concrete-additive mixture shells

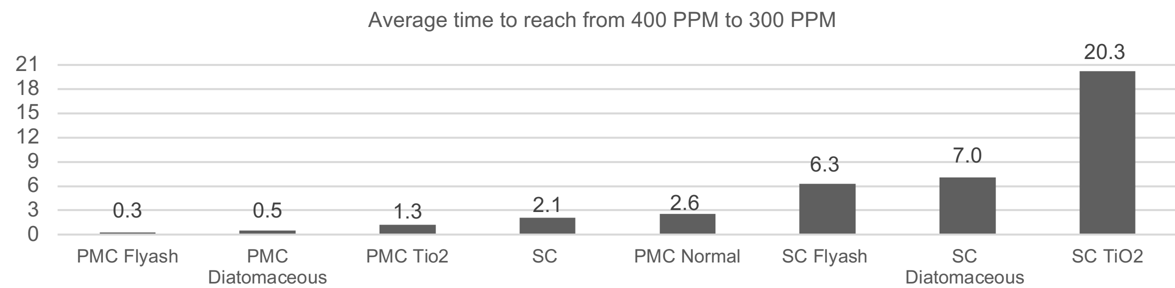


Figure 10 : Bar graph in ascending order of time to drop from 400 PPM to 300 PPM CO_2 levels for the different concrete-additive mixture shells

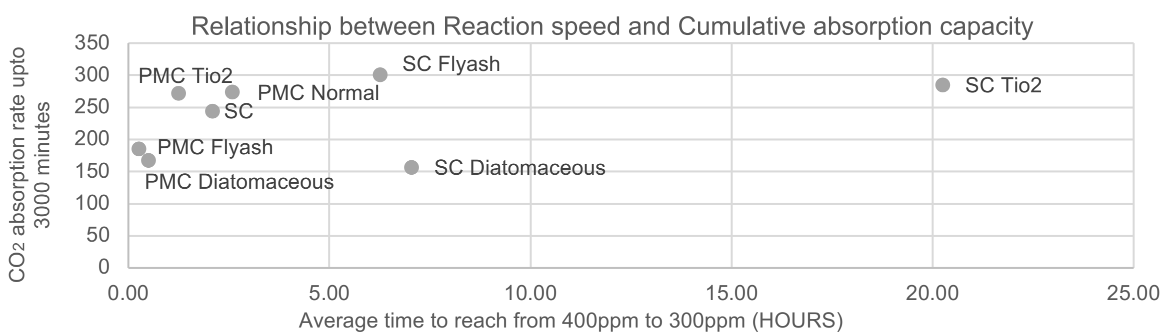


Figure 11 : The bubble scatter plot illustrates the relationship between the reaction speed (represented as the average time required for CO_2 concentration to decrease from 400 ppm to 300 ppm) and the CO_2 absorption rate (measured over 3000 minutes) for different concrete mix types

4.0 CONCLUSION

This research demonstrates that integrating digitally knitted formwork with carbon-absorbing concrete composites provides a promising pathway toward sustainable, adaptive, and materially efficient construction methods. Through combined experimental and computational analysis, the study established that knitted fabric structures exhibit predictable anisotropic deformation under hydrostatic pressure, which can be modeled using nonlinear finite-element simulation with reasonable accuracy. The close correlation between simulated and fabricated shell geometry validates the digital workflow as a reliable design tool for fabric-formed concrete. The results from CO_2 absorption experiments confirm that additive-rich concrete, particularly mixtures incorporating fly ash and titanium dioxide, significantly enhances carbon sequestration compared to control specimens. Moreover, the use of polymer-modified matrices accelerated CO_2 uptake, suggesting a viable route for improving short-term reactivity without compromising long-term durability. When paired with high-surface-area geometries generated through knitted formwork, these materials create active, surface-responsive shells that are capable of capturing measurable amounts of atmospheric carbon after curing. Beyond environmental performance, the study underscores the architectural and logistical advantages of fabric formwork. Its lightweight, reusable, and digitally customizable nature reduces material waste, transportation demands, and formwork costs. The visual and tactile richness of knitted textures further extends the aesthetic and expressive potential of concrete, merging softness and structure into a single sustainable design language. Figure 12 captures the sculptural quality and tactile texture of each shell, where the knitted loops translate into rhythmic surface patterns in hardened concrete.

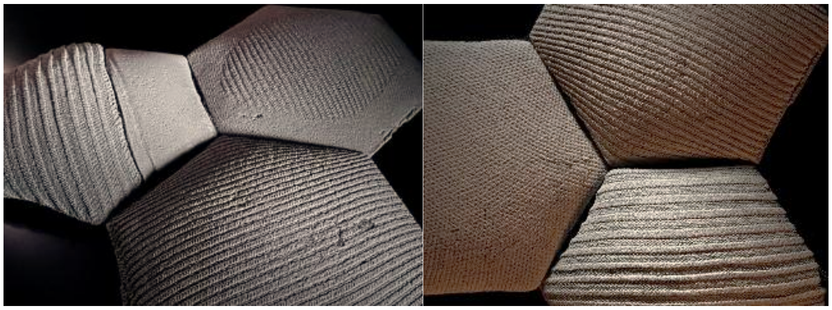


Figure 12 : Fabric-formed concrete shells for display.

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REFERENCES

Anderson, Jane, and Alice Moncaster. 2020. “Embodied Carbon of Concrete in Buildings, Part 1: Analysis of Published EPD.” *Buildings and Cities* 1 (1): 198–217. https://doi.org/10.5334/bc.59.

Araujo, M De. 2011. *Weft-Knitted Structures for Industrial Applications*.

Cooke, B. 2011. *The Physical Properties of Weft Knitted Structures*.

D13 Committee. n.d. *Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)*. ASTM International. https://doi.org/10.1520/D5034-21R25.

Fang, Rosemarie. 2009. “The Design and Construction of Fabric Structures.” MIT.

Kipsanai, Janet J., Paul M. Wambua, Saul S. Namango, and Sofiane Amziane. 2022. “A Review on the Incorporation of Diatomaceous Earth as a Geopolymer-Based Concrete Building Resource.” *Materials* 15 (20): 7130. https://doi.org/10.3390/ma15207130.

Li, Chaohui, Prajal Pradhan, Guoqian Chen, Jürgen P. Kropp, and Hans Joachim Schellnhuber. 2025. “Carbon Footprint of the Construction Sector Is Projected to Double by 2050 Globally.” *Communications Earth & Environment* 6 (1): 831. https://doi.org/10.1038/s43247-025-02840-x.

Orr, John. 2019. “Flexible Formwork for Concrete Structures.” Apollo - University of Cambridge Repository. https://doi.org/10.17863/CAM.39157.

Popescu, M., L. Reiter, A. Liew, T. Van Mele, R.J. Flatt, and P. Block. 2018. “Building in Concrete with an Ultra-Lightweight Knitted Stay-in-Place Formwork: Prototype of a Concrete Shell Bridge.” *Structures* 14 (June): 322–32. https://doi.org/10.1016/j.istruc.2018.03.001.

The Institute of Paper Chemistry, G. A. Baum, C. C. Habeger, The Institute of Paper Chemistry, E. H. Fleischman, and The Institute of Paper Chemistry. 1981. “Measurement of the Orthotropic Elastic Constants of Paper.” *Trans. of the VIIth Fund. Res. Symp. Cambridge, 1981*, 453–78. https://doi.org/10.15376/frc.1981.1.453.

Valentini, Luca. 2023. “Sustainable Sourcing of Raw Materials for the Built Environment.” *Materials Today: Proceedings*, August, S2214785323041780. https://doi.org/10.1016/j.matpr.2023.07.308.

Vassiliadis, Savvas G, Argyro E Kallivretaki, and Christopher G Provatidis. 2007. “Geometrical Modelling of Plain Weft Knitted Fabrics.” *INDIAN J. FIBRE TEXT. RES.*

Veenendaal, Diederik, and Philippe Block. 2014. “Design Process for Prototype Concrete Shells Using a Hybrid Cable-Net and Fabric Formwork.” *Engineering Structures* 75 (September): 39–50. https://doi.org/10.1016/j.engstruct.2014.05.036.

West, Mark. 2017. *The Fabric Formwork Book: Methods for Building New Architectural and Structural Forms in Concrete*. Routledge. https://doi.org/10.4324/9781315675022.